

HW3-Q1. Suppose the common source stage of the figure below is to provide an output swing from 1V to 2.5V. Assuming $(W/L) = 50/0.5$, $R_D = 2\text{k}\Omega$, and $\lambda = 0$.

- Find the input voltages that yield $V_{\text{out}} = 1\text{V}$ and $V_{\text{out}} = 2.5\text{V}$.
- Find the drain current and transconductance of M_1 for both cases.
- How much does the small signal gain, $g_m \times R_D$, vary as the output goes from 1V to 2.5V? (variation of small signal gain can be viewed as nonlinearity).

